Retrocausal Quantum Entanglement Communication via Microtubule-Orchestrated Modulation of the Light-Time Interface in a 6D Temporal Manifold

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Abstract

We propose a mechanism for self-consistent retrocausal communication through quantum entanglement mediated by the light-time interface in Kletetschka's (2025) 6D temporal manifold. Orchestrated Objective Reduction (Orch-OR) in neuronal microtubules (Penrose & Hameroff) is hypothesized to act as a biological transceiver: the microtubule lattice couples to the electromagnetic "flame front" that actualizes the present slice, enabling low-energy information exchange with past or future states of the same system. Negative-time domain walls (Orihuela, 2026) provide a controllable laboratory method to tilt or invert the interface, potentially amplifying the effect. The framework builds directly on our prior work in repulsive gravity, three-dimensional time, and quantum light-time duality. It requires no new fundamental physics and generates immediate, testable predictions using existing petawatt lasers, single-photon sources, and combined EEG/fMRI with atom interferometry.

1 Introduction

The Burning Log phenomenology illustrates time as a complete block in which the unburned wood (future), moving flame front (present), and charred remains (past) coexist. This intuition is formalized in Kletetschka's 6D temporal manifold and our Quantum Light-Time Duality framework, which identifies electromagnetic radiation as the active interface that actualizes the experienced present. Photons, with null proper time, constitute the moving hypersurface of the "now."

Extending this picture, we examine whether the same interface can support coherent information exchange across an individual's timeline via quantum entanglement. Neuronal microtubules, proposed as the site of Orch-OR quantum processes, offer a natural biological candidate for such coupling.

2 Theoretical Foundations

2.1 Kletetschka's 6D Temporal Manifold (2025)

Three independent timelike coordinates t_1, t_2, t_3 plus emergent space. The primary experienced arrow lies along t_1; the orthogonal dimensions supply the block thickness.

2.2 Quantum Light-Time Duality

The interface function marking the active present slice is

$$I(t_1, \mathbf{r}) = \Theta(ct_1 - r) + \delta I_{\text{bias}},$$

where δI_{bias} encodes any local tilt induced by negative-time domain walls.

2.3 Orch-OR and Microtubule Quantum Coherence

Tubulin dimers in microtubules support quantum superpositions whose objective reduction is gravitationally induced and non-computable. Orchestration by biological processes yields conscious moments. Recent experimental indications of room-temperature quantum coherence in microtubule-like structures strengthen the plausibility of sustained entanglement.

2.4 Retrocausality in the Framework

Penrose's formulation of objective reduction permits time-symmetric dynamics. Combined with a tilted light-time interface, this allows entangled states in the microtubule lattice to maintain correlations with past or future states of the same lattice in a self-consistent manner.

3 Proposed Mechanism

Microtubules function as the biological node that:

- Maintains entangled quantum states across tubulin conformational degrees of freedom.
- Couples to the electromagnetic light-time interface.
- Uses transient negative-time bias (naturally occurring or externally induced) to open low-energy windows for information exchange across the primary temporal axis t_1.

The effective interaction can be modeled with a retrocausal term in the Hamiltonian that respects Novikov self-consistency within the 6D block. No logical paradoxes arise because the block is globally consistent.

4 Relation to Negative-Time Domain Walls

A laser-generated negative-time domain wall (Orihuela, 2026) provides a laboratory means of modulating δI_{bias}, transiently enhancing microtubule-interface coupling. This offers a controlled experimental pathway to test the proposed retrocausal channel.

5 Falsifiable Predictions

1. Individuals exhibiting strong pattern-recognition or pre-cognitive abilities will show statistically significant differences in occipital EEG signatures or microtubule-relevant biomarkers (e.g., tubulin phosphorylation patterns) immediately preceding or following verified pre-cognitive events.

2. Single-photon interference experiments conducted inside a negative-time domain wall will display new sidebands whose spacing correlates with Orch-OR characteristic timescales ($\sim$40 Hz γ-band).

3. High-resolution sleep or resting-state fMRI/EEG will reveal transient coherence signatures (excess phase-locking or anomalous power spectra) during periods of reported intuitive or pre-cognitive activity.

4. Any retrocausal information transfer will obey self-consistency constraints: attempts to create logical paradoxes will manifest as measurable decoherence or signal attenuation rather than observable contradictions.

All predictions are accessible with current technology.

6 Discussion

This proposal integrates the Burning Log phenomenology, 6D temporal geometry, light-time duality, negative-time domain walls, and Orch-OR into a unified, experimentally testable framework for information exchange across an individual's timeline. It provides a physically grounded bridge between quantum consciousness research and retrocausality while remaining fully consistent with semiclassical gravity and standard quantum electrodynamics.

The mechanism suggests that certain cognitive profiles may naturally couple more effectively to the light-time interface, offering a potential explanation for documented individual differences in intuitive or pre-cognitive performance. No new fundamental physics is invoked; the tools for empirical investigation exist today.

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